

Name: \_\_\_\_\_ / 25

Please circle your section:      901 (Gossell)      902 (Gossell)      903 (Graves-McCleary)

No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. [12 points] Solve the following limits. Show your work and use correct limit notation.

a.  $\lim_{x \rightarrow 2} \frac{x^3 - 2x^2}{6 - 3x}$

b.  $\lim_{x \rightarrow 1} \frac{\frac{x}{2} - \frac{2}{x}}{2x - 4}$

c.  $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$

2. [8 points] Consider the following piecewise function:

$$g(x) = \begin{cases} \frac{x^2 - 4}{x + 2}, & x < -2 \\ \sqrt{4 - x^2}, & -2 \leq x \leq 2 \\ x^2 - 4, & x > 2 \end{cases}$$

a. Is  $g(x)$  **continuous** at  $x = -2$ ? Justify your answer **using limits**.

b. Is  $g(x)$  **continuous** at  $x = 2$ ? Justify your answer **using limits**.

3. [5 points] Use the Intermediate Value Theorem to show that the graph of  $f(x) = 3^x - x - 20$  passes through the  $x$ -axis between  $x = 2$  and  $x = 3$ .